

# Data analytics Project:

TOPIC: AI Job Market  
Trends.

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# INTRODUCTION:

- ▶ *This project explores the AI job market using Python data analysis and data visualization techniques. It aims to uncover patterns in AI job salaries, demand, Experience ,Job role and skill trends.*



Tech



Healthcare



Finance



Manufacturing



Retail

# OBJECTIVE:

- ▶ • **Analyze AI job salaries across regions**
- ▶ • **Identify top in-demand AI skills**
- ▶ • **Visualize growth and opportunities in AI job sectors**



# DATASET OVERVIEW:

- ▶ Dataset: AI Job Market Trends 2024-2025
- ▶ • Rows: 1000 job postings
- ▶ • Columns: Job\_ID, Company, Location, Salary, Skills, Experience Level
- ▶ • Data Source: Public AI job listings & Kaggle dataset

# DATA CLEANING & PREPARATION:

- ▶ • *Removed duplicates and missing salary entries*
- ▶ • *Normalized job titles and skill names*
- ▶ • *Converted salary data to consistent currency (USD)*
- ▶ • *Filtered AI-related roles only*

# TOOLS & LIBRARIES USED:

- ▶ • Pandas – Data handling and preprocessing
- ▶ • Matplotlib – Static data visualization
- ▶ • Seaborn – Enhanced statistical plots
- ▶ • NumPy – Numerical operations

seaborn

Pandas



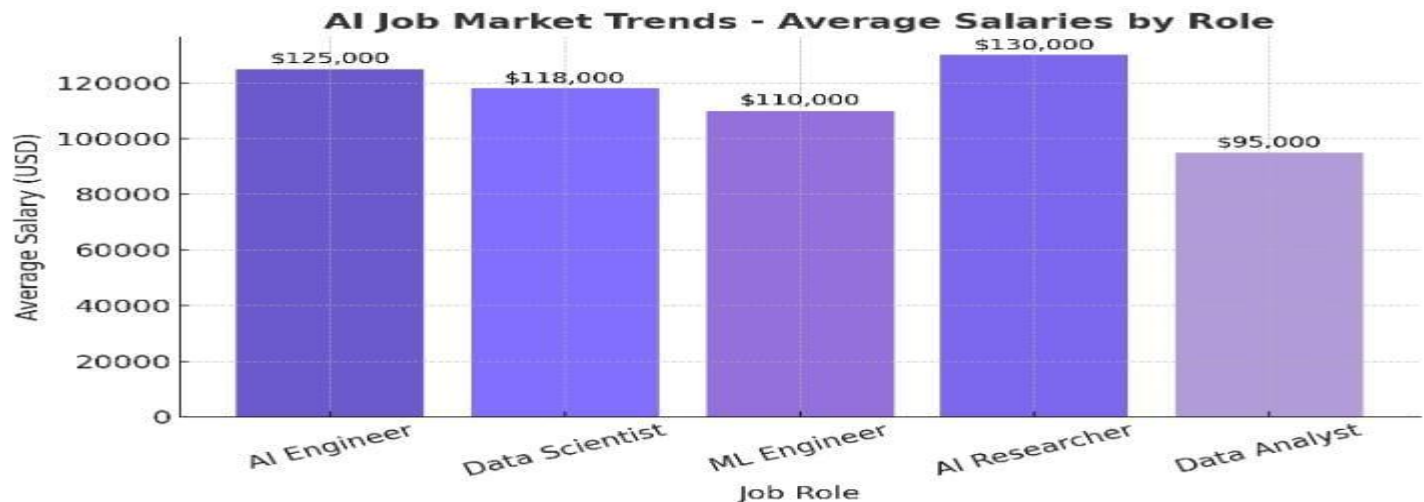
matplotlib



# SALARY TREND

## VISUALIZATION:

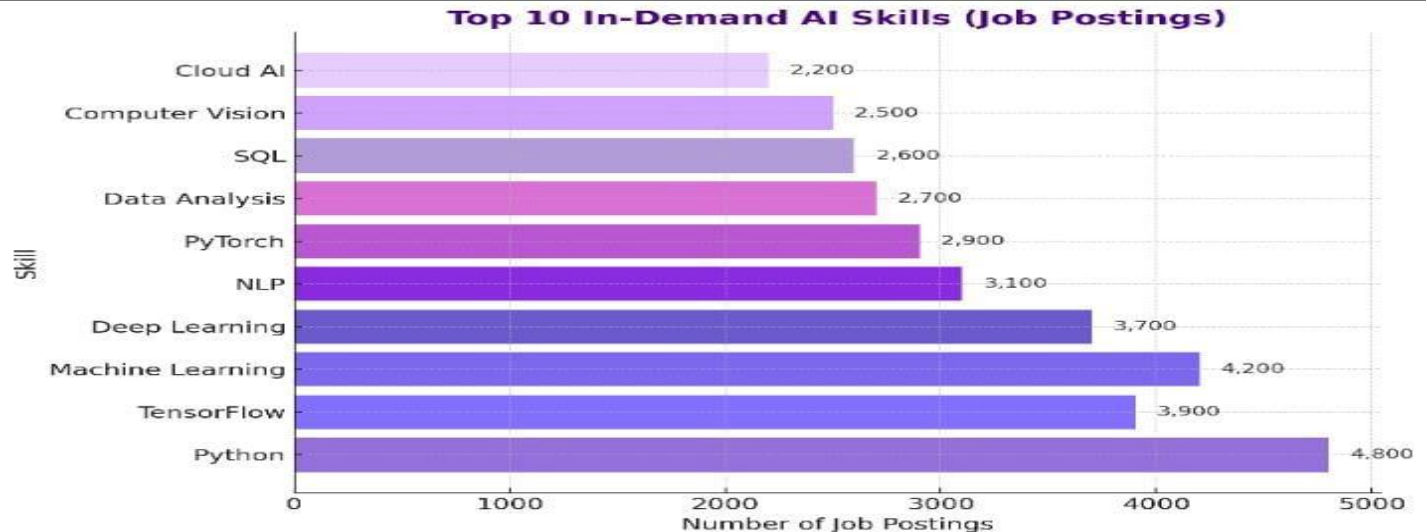
- ▶ Visualized average AI job salaries using bar and line plots.
- ▶ [Example code:](#)
- ▶ `import matplotlib.pyplot as plt`
- ▶ `import seaborn as sns`
- ▶ `sns.barplot(x='Job Title', y='Salary', data=df)`
- ▶ `plt.title('Average Salary by AI Job Role')`
- ▶ `plt.show()`



# SKILL DEMAND

## VISUALIZATION:

- ▶ Visualized skill frequency to identify most required AI skills.
- ▶ Example code:
- ▶ `skill_counts = df['Skills'].value_counts().head(10)`
- ▶ `sns.barplot(x=skill_counts.values, y=skill_counts.index)`
- ▶ `plt.title('Top 10 In-Demand AI Skills')`
- ▶ `plt.show()`



# REGIONAL & INDUSTRY TRENDS:

- ▶ Mapped job demand by region and industry.
- ▶ Example visualization: heatmap showing AI job distribution by region.
- ▶ `sns.heatmap(job_data, cmap='coolwarm')`



## INSIGHTS & KEY FINDINGS:

- ▶ • AI Engineers and Data Scientists are top-paid roles.
- ▶ • Python, TensorFlow, and NLP skills are most in-demand.
- ▶ • The U.S., U.K., and India lead in AI job openings.
- ▶ • Increasing demand for AI skills in finance and healthcare sectors.

# Challenges & Solutions

- ▶ • Data inconsistency and duplicates – handled with Pandas cleaning.
- ▶ • Visual clutter – solved by using Seaborn's advanced plots.
- ▶ • Limited data coverage – future inclusion of more job portals planned.

# Conclusion & Future Scope

- ▶ **This project demonstrates how Python can effectively visualize complex job market data.**
- ▶ **Future work:**
  - ▶ **• Predict AI salary trends using ML models**
  - ▶ **• Interactive dashboards with Plotly or Power BI**
  - ▶ **• Broader dataset integration for global trends.**

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***THANK YOU***

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